

Angadh Verma

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EDUCATION

McGill University

Bachelor of Arts in Computer Science and Statistics

- Dean's Honor List, GPA: 3.85/4.00.

Montréal, Canada

Jan. 2022 – Apr. 2026

EXPERIENCE

Software Developer Intern – Machine Learning

May 2025 – Aug. 2025

Retail Realm Distribution

Sonoma, California (Remote)

- Designed and deployed a fully local **retrieval-augmented generation** (RAG) system using LangGraph-based **agentic workflows** to help tech-support teams query historical ticket data, reducing average ticket resolution time by 35%.
- Built an end-to-end data pipeline with **Azure** Blob Storage, SQL Server and **PySpark** to populate a **ChromaDB** vector database, modeling issue-resolution relationships to improve retrieval quality and reduce hallucinations.

Statistical Modelling Research Intern

Jun. 2024 – Aug. 2024

Indian Institute of Technology

New Delhi, India

- Automated a data-processing pipeline in **Google Earth Engine** and Python for **geospatial satellite imagery** used to study the impact of farm ponds on crop health (NDVI) and water security (NDMI).
- Applied Differences-in-Differences and Double ML based **causal analysis** methods to estimate treatment effects across crop-yield, soil-moisture, and vegetation outcomes, contributing to a paper presented at ACM COMPASS.

PROJECTS

BookFlow | *Next.js, FastAPI, MySQL, Auth0, LangChain*

- Built a full-stack library management system with **role-based access** control via Auth0 JWT validation, a multi-state borrowing workflow, and an **AI search assistant** with LangChain and heuristic fallback handling.
- Integrated an **AI-powered metadata enrichment pipeline** that auto-populates book genre, summary, and cover images using OpenAI-compatible LLMs with OpenLibrary as a structured fallback, deployed across **Vercel, Render, and Aiven**.

InsideRadar | *Polars, FastAPI, SQLAlchemy, XGBoost*

- Built a full-stack web application that streams and curates live **Polymarket trading data** to rank potentially insider-driven bets using residual analysis and outlier detection on an XGBoost model.
- Implemented the data-science pipeline and backend architecture using Polymarket APIs, Polars for efficient feature engineering, FastAPI and SQLAlchemy for secure API endpoints, and Auth0 for authentication.

Hate Speech Data Pipeline | *ETL, LLMs*

- Developed an ETL framework for aligning hate-speech datasets and used an OpenAI **GPT-3.5 turbo** for **semantic schema matching** across heterogeneous annotation labels, improving reproducibility for future research in the Mila Complex Data Lab.

Statistical Consulting | *ArcGIS, Geospatial Analysis, Flight Data*

- Synechron**: Conducted geospatial analysis using ArcGIS to **visualize bank branch** networks, perform statistical **network analysis**, and inform strategic ATM placement decisions for senior management.
- IATA**: Processed high-throughput FlightRadar24 data to create a geometry-based aircraft collision-detection algorithm, providing a secondary safety layer when onboard TCAS data is unavailable.

Intelligent Credit Risk Portfolio Dashboard | *Scikit-Learn, Power BI*

- Designed an interactive dashboard with DAX-based financial KPIs on an end-to-end predictive analytics pipeline in Python and Scikit-Learn to train a Random Forest model on more than 100,000 LendingClub loan records.

LEADERSHIP / EXTRACURRICULAR

Software Designer | *McGill Robotics Drone Team*

Sept. 2025 – Present

- Developing embedded algorithms for real-time mapping and object detection on an autonomous drone.

Co-coordinator | *The Yellow Door Org*

Sept. 2025 – Present

- Manage and cook for a weekly community kitchen serving underprivileged students and seniors.

TECHNICAL SKILLS

Languages: Python, SQL, Java, R, Javascript

Data & ML Tools: SQL Server, PySpark, Databricks, Polars, Pandas, Scikit-Learn, PyTorch, Keras, LangChain, LangGraph, Ollama, Vector DB, Power BI, Tableau

Deployment: Git, Docker, Vercel, Render, Streamlit